Design Proposal for Interferometry Using Mach-Zehnder Interferometers and Ring Resonators

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May 8, 2026

1 Introduction

Silicon photonics is a growing field that aims to replicate the vesatility and applications of free-space or fibre-based optics, using mature silicon fabrication processes, to miniaturise systems: an optical set-up that may require an entire optical bench in free space could be achieved on a silicon chip centimetres or milimetres in size. This is achieved through micrometre and nanometre-scale waveguide-based passive components, such as beamsplitters, crossings and couplers, in concert with active components such as phase changers, and on-chip light sources and detectors. In this article, I propose three different designs for performing interferometry: a Mach-Zehnder interferometer (MZI), a coupled ring resonator-based optical reflector, and a ring resonator-based demultiplexer (demux). These three devices will allow calculation of the group index in a silicon waveguide.

2 Theory

In this section I will outline the theory of the MZI and the ring resonator, and show how group index can be calculated through measurments of these devices.

2.1 Mach-Zehnder interferometer

Simply put, a MZI is two coupled 2x2 splitters. The incident light is split at the first splitter, and then interferes at the second splitter, changing the amplitude at each output. By introducing a path length difference in the intermediate arms of the MZI, creating a so-called "unbalanced" MZI, the fraction of the transmitted light from each output changes as a function of the phase shift due to the path length difference.

The electric field of the light transmitted to the intermediate arms of the MZI by the initial splitter is given by: $E_1 = E_2 = \frac{E_i}{\sqrt{2}}$. The light propagates (with propagation constant $\beta = 2\pi n_{eff}/\lambda$) down waveguides with path length L_1 and L_2. These meet at

the second splitter. At the input to the second splitter, the fields are given by:

$$\begin{array}{rcl} E_1 & \rightarrow & E_1 e^{-i\beta L_1} \\ E_2 & \rightarrow & E_2 e^{-i\beta L_2} \end{array}$$

The field after the splitter on one of the outputs is

$$E_o = \frac{E_1 + E_2}{2}$$

and the intensity is

$$I_o = \frac{(E_1^* E_1 + E_2^* E_2 + E_1^* E_2 + E_2^* E_1)}{4}$$

The $E_i^* E_j$ terms with $i = j$ give a contribution of $I_i/2$, whereas the interference terms where $i \neq j$ introduce a sinusoidal term dependent on the path length difference $\Delta L = L_1 - L_2$. This results in output intensity:

$$\begin{array}{rcl} I_o & = & \frac{I_i}{2}(1 + \cos(\beta \Delta L) \\ & \propto & (1 + \cos(2\pi n_{eff} \Delta L/\lambda) \end{array}$$

Thus by sweeping wavelength, the intensity will oscillate between maximum and zero. The peak-to-peak distance ΔL is known as the free spectral range (FSR). The group index can be calculated from the FSR.

2.2 Ring resonator

A ring resonator is a circular waveguide that acts as an optical cavity. Light is fed into it from an adjacent waveguide via the evanescent field, and round-trip interference for light of different wavelengths means that only certain modes are supported in the resonator. The optical path length of a ring resonator is given by $2\pi r n_{eff}$ where r is the radius of the ring. For resonance of a given mode, the optical path length must be equal to an integer number of wavelengths: $m\lambda_m$. For a given size of ring resonator, the FSR is the difference in wavelength between modes m and $m \pm 1$, i.e. the modes where the ring supports m and $m \pm 1$ wavelengths. This corresponds to a phase difference of 2π over the full ring round trip, so the FSR can be calculated similarly to the MZI: $\Delta\beta L = 2\pi, \beta = 2\pi n(\lambda)/\lambda, L = 2\pi r$. By expanding $\Delta\beta$ as a Taylor series, the FSR is given as

$$\Delta\lambda = \frac{\lambda^2}{n_g L} = \frac{\lambda^2}{2\pi r n_g}$$

where the group index $n_g = n(\lambda) - \lambda \frac{\partial n}{\partial \lambda}$.

2.2.1 Optical reflector

This device is given by two identical ring resonators in series along a waveguide that are in turn coupled together. Modes that couple into the first ring in the anticlockwise direction can in turn couple into the second ring in the clockwise direction, and then

couple into the base waveguide propagating backwards. In turn, modes that couple into the second ring resonator from the waveguide can couple into the second ring and then couple back into the waveguide, also propagating backwards. Thus the device acts as a wavelength-selective filter/mirror, reflecting a portion of the incident light back along the waveguide, resulting in a reduction in the transmitted power at the ring modes. The FSR can be extracted from the transmitted spectrum as a series of dips.

2.2.2 Demultiplexer

This device is made up of a series of cascaded ring resonators that are coupled to two waveguides at opposite sides of the ring. Resonant modes of the first ring are coupled to the second waveguide output, allowing collection of light of those resonant modes. By changing the radii of the subsequent ring resonators, successive sets of modes can be picked off the incident light and coupled out to detectors. This allows a number of ring resonators to be measured from one input, each with its own FSR, and thus each allowing another calculation of the group index.

3 Modelling and simulation

Modelling of the waveguides and MZIs was performed using Lumerical MODE and INTERCONNECT. Modelling of ring resonators performed using the 2.5D varFDTD mode in Lumerical MODE - full 3D FDTD simulations were not performed due to the size of the structures.

3.1 TE waveguide mode

The waveguide geometry was standard for 1.55μm wavelength: 0.5 μm width and 0.22 μm thickness. Mode calculations were performed using MODE's fiite difference eigenmode solver and showed fundamental mode indices of 2.448 for TE amd 1.76 for TM at 1.55 μm. A frequency sweep from 1.5 to 1.6 microns show a strongly linear response for the real part between 2.50 and 2.39. The group index over the same range rises from 4.17 to 4.28.

MODE simulations were also perfomed for bent waveguides with radius around 20 microns - similar to those of the ring resonators (assuming m of around 200). The values for effective and group index from these simulation show very good agreement with those of the straight waveguides.

3.2 MZI

Three designs for MZIs are included in the layout, with different ΔL values, outlined in Table. 1. The MZI circuits from the .gds file were simulated in Lumerical INTERCONNECT via KLayout, and mean group index values were calculated from the FSRs.

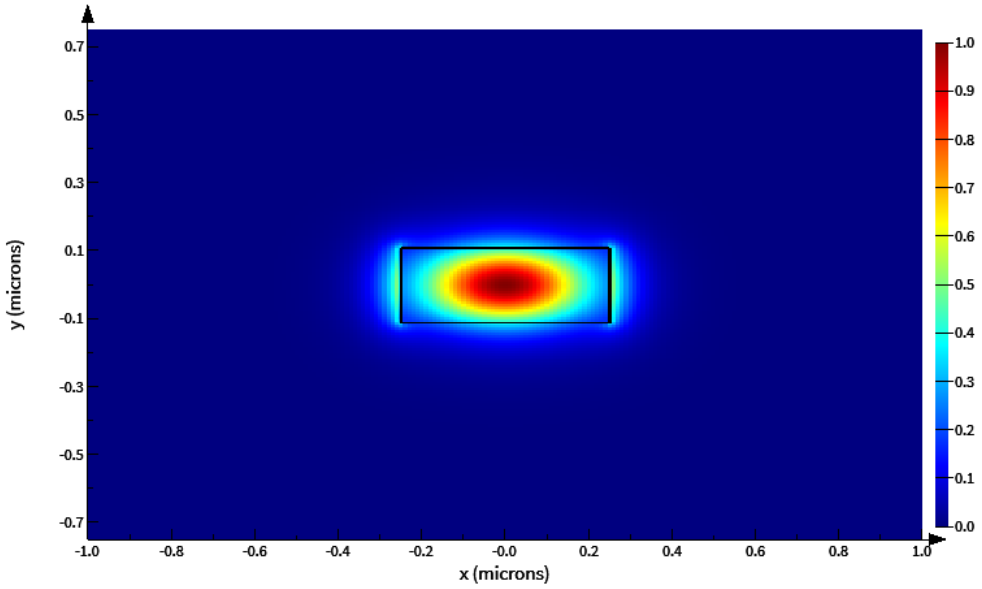
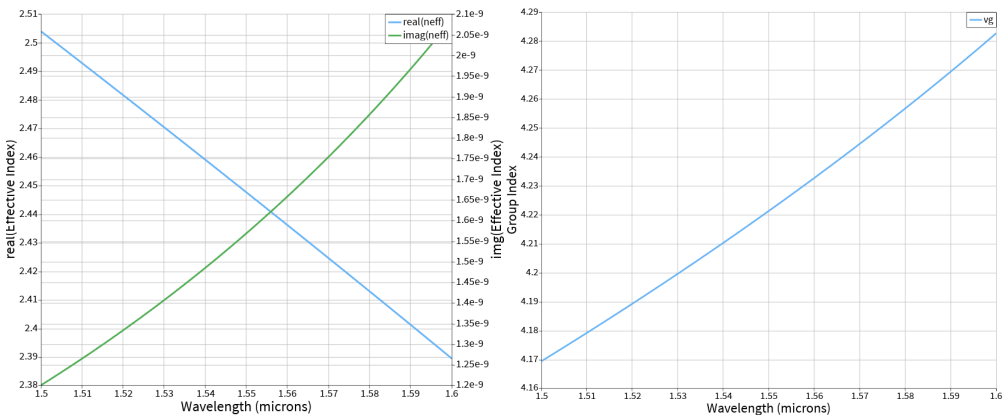


Figure 1: Fundamental TE mode in waveguide



(a) (b)

Figure 2: (a) plot of effective index (b) plot of group index. Both from MODE simulation.

3.3 Dual-bus ring resonator

This component is used in the optical demultiplexer - see Fig. **??**. The dual-bus ring is coupled to two waveguides at opposite sides of the ring.[1] Light at the correct wavelength,

[1]The names of the ports are chosen due to their relationship to the input waveguide: 'through' carries light that does not couple to the ring. Light that couples to the ring from 'input' can then leave the ring and exit through 'drop' - light is dropped from the through waveguide. If light were incident at 'add' and coupled to the ring, it would be 'added' to the input-through waveguide.

MZI	ΔL (μm)	simulated n_g
1	244.4	4.22 ± 0.08
2	103.5	4.26 ± 0.06
3	396.3	4.21 ± 0.03

Table 1: Table of MZI path lengths and simulated group indices (n_g).

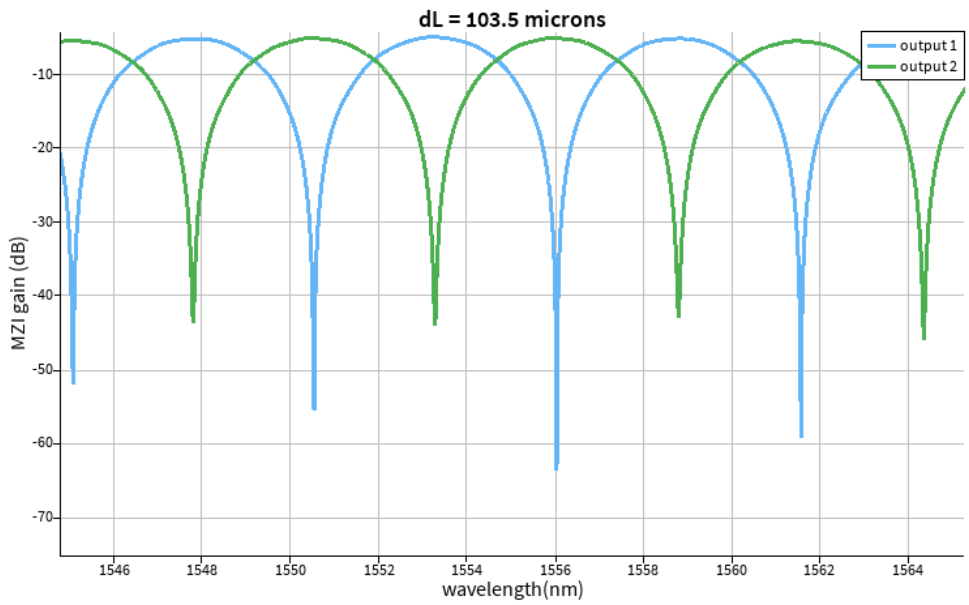


Figure 3: Plot of gain to both outputs of MZI circuit. Distance between peaks is FSR.

given by the resonance relationship above, can couple into the ring and circulate, and in turn couple to the second waveguide and exit through 'drop'. The coupling is determined by the evanescent field established by the waveguide - in simple terms, the closer the waveguide is to the ring, and the longer that the waveguide is close to the ring, the higher the degree of coupling.

In my designs, I have included two ring-waveguide gaps: 100nm and 200nm. Performing varFDTD simulations, the difference in coupling strength can be seen for the different gaps: see Fig. **??**. The smaller gap clearly allows more light to couple into the ring. This is further demonstrated in plots of the power at different points on the ring, as seen in Fig. **??**. A smaller gap results in, at resonance, lower power to the through port, equal power at opposite points in the ring, and greater power to the drop port.

Three ring radii are included in the multiplexer, corresponding to three sets of resonant modes. The radii are calculated to support three chosen wavelengths with $m = 200$. These are given in Table. **??**.

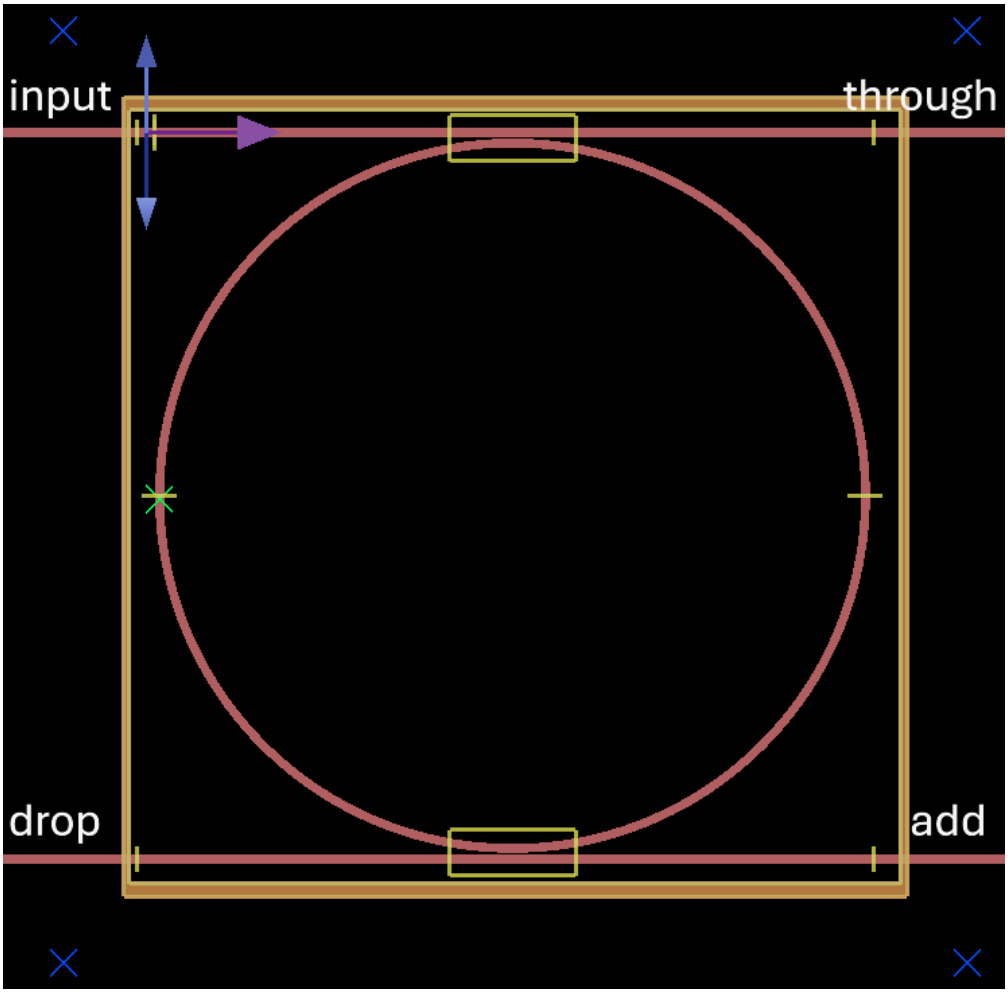


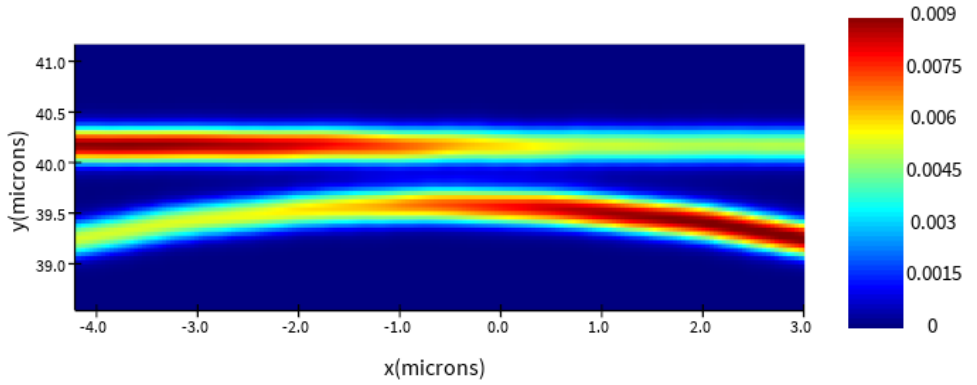
Figure 4: Lumerical MODE view of dual-bus ring. Input/output ports are labelled.

target wavelength (μm)	radius (μm)
1.50	19.09
1.55	20.15
1.60	21.32

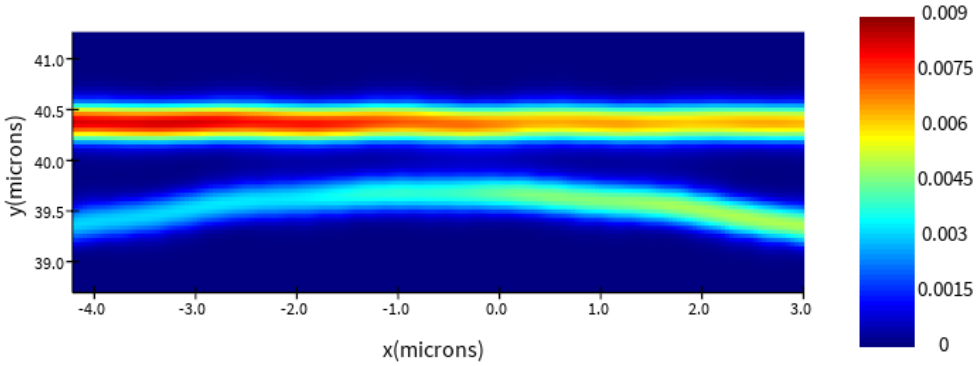
Table 2: Table of target resonant wavelengths with $m = 200$ and calculated ring resonator radii.

3.4 Optical reflector

The key component in the optical refelctor is the coupled ring resonator: see Fig. **??**. Light on resonance travels an S-shaped path through the device, exiting back the way it came. varFDTD simulations how this behaviour. Much like the dual-bus ring resonators, two waveguide-ring gaps are chosen: 100nm and 200nm. These affect the efficiency of



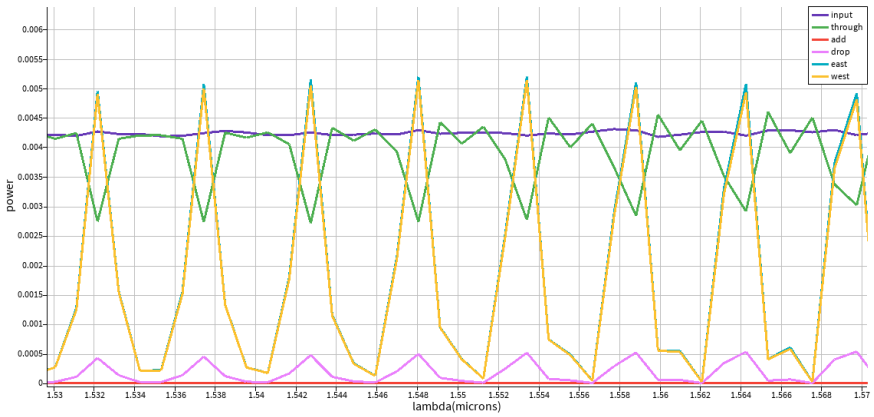
(a)



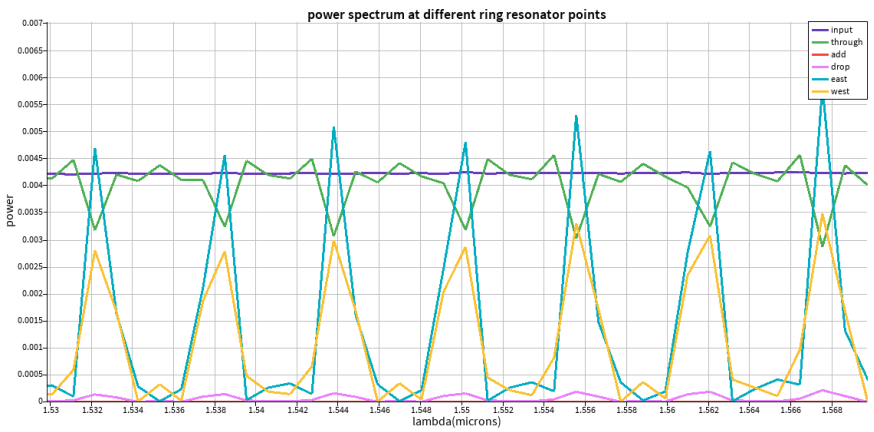
(b)

Figure 5: FDTD simulation detail of ring resonator coupler for gasp of (a) 100nm and (b) 200nm. Changing the gap gives a small ($\tilde{2}$nm) shift to the resonant wavelength: this may be due to interference at the couplers affecting the optical path length in the ring.

the reflection in the expected way: see Fig. **??**. The ring radii are chosen as 20.15 mum, with target wavelength 1.55 mum.

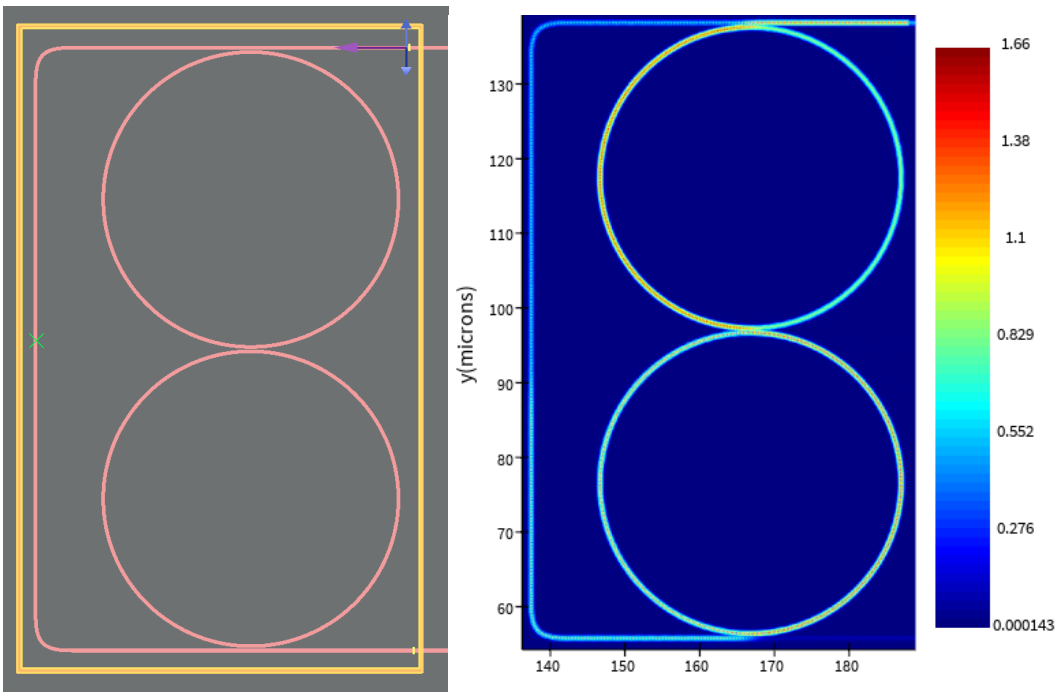


(a)



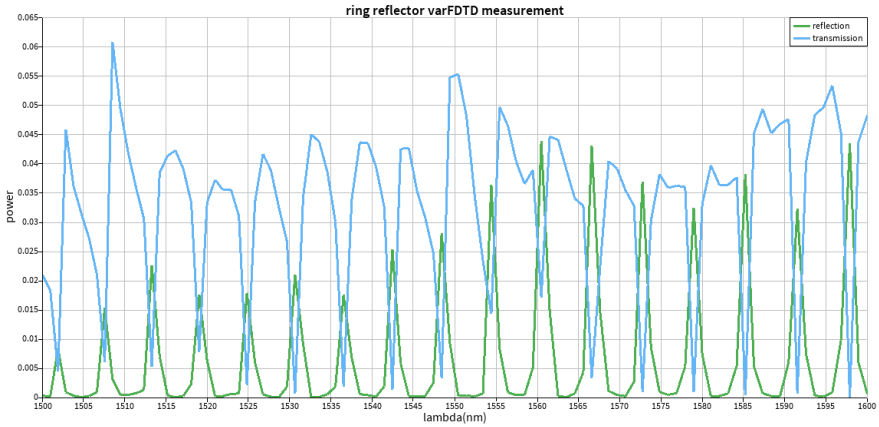
(b)

Figure 6: Power spectra at different monitors in the ring reflector with gap of (a) 100nm and (b) 200nm. 'East' and 'west' refer to monitors at the leftmost and rightmost parts of the ring, as seen in Fig. **??**.

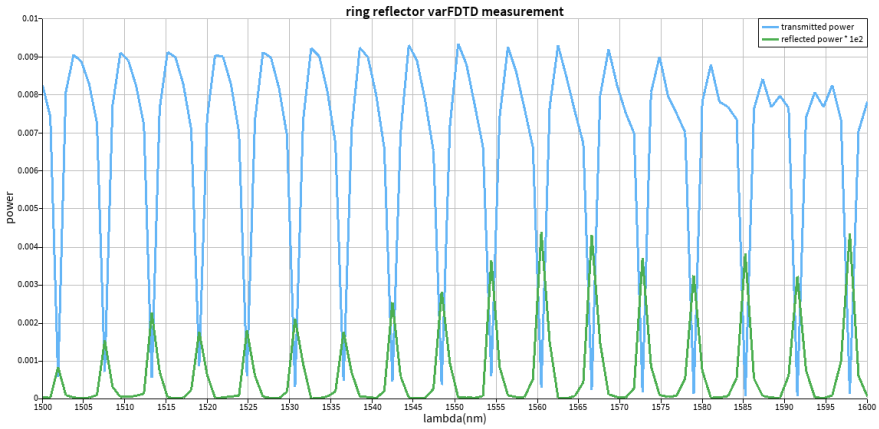


(a) (b)

Figure 7: (a) structure of coupled ring resonator in Lumerical MODE. (b) varFDTD simulated electric field in the device close to ring resonance. Note significantly reduced field at output (bottom right).



(a)



(b)

Figure 8: Transmission and reflection spectra for coupled rings with gap (a) 100nm and (b) 200nm. Note in (b): the reflection signal has been artifically amplified by a factor of 100 to be visible on the plot.